

Prob 1

$$F = [3y, 2x, 4z]$$

HW #5: Line Integrals

$$(i) \quad \vec{r}(t) = [t, t, t] \quad 0 \leq t \leq 1$$

$$\vec{r}'(t) = [1, 1, 1]$$

$$F(\vec{r}(t)) = [3t, 2t, 4t]$$

$$F(\vec{r}(t)) \cdot \vec{r}'(t) = 3t + 2t + 4t = 9t$$

$$\int_{C_1} F \cdot d\vec{r} = \int_0^1 9t \, dt = \left. \frac{9}{2} t^2 \right|_0^1 = \frac{9}{2}$$

$$(ii) \quad \vec{r}(t) = [\sqrt{t}, t, t^2] \quad 0 \leq t \leq 1$$

$$\vec{r}'(t) = \left[\frac{1}{2} t^{-1/2}, 1, 2t \right]$$

$$F(\vec{r}(t)) = [3t, 2\sqrt{t}, 4t^2]$$

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = \frac{3}{2} t^{1/2} + 2t^{1/2} + 8t^3 = \frac{7}{2} t^{1/2} + 8t^3$$

$$\begin{aligned} \int_{C_2} F \cdot d\vec{r} &= \int_0^1 \left(\frac{7}{2} t^{1/2} + 8t^3 \right) dt = \left. \frac{7}{2} \cdot \frac{2}{3} t^{3/2} + \frac{8}{4} t^4 \right|_0^1 \\ &= \frac{7}{3} + 2 = \frac{13}{3} \end{aligned}$$

$$(iii) \quad \vec{r}_1(t) = [t, t, 0] \quad 0 \leq t \leq 1$$

$$\vec{r}_1'(t) = [1, 1, 0]$$

$$\vec{F}(\vec{r}_1(t)) = [3t, 2t, 0]$$

$$\vec{F}(\vec{r}_1(t)) \cdot \vec{r}_1'(t) = 3t + 2t = 5t$$

$$\begin{aligned} \int_{C_3} \vec{F} \cdot d\vec{r} &= \int_0^1 5t \, dt = \left. \frac{5}{2} t^2 \right|_0^1 \\ &= \frac{5}{2} \end{aligned}$$

$$\begin{aligned} \vec{r}_2(t) &= [0, 0, t] + [1, 1, 0] \\ &= [1, 1, t] \end{aligned}$$

$$\vec{r}_2'(t) = [0, 0, 1]$$

$$\vec{F}(\vec{r}_2(t)) = [3, 2, 4t]$$

$$\vec{F}(\vec{r}_2(t)) \cdot \vec{r}_2'(t) = 4t$$

$$\begin{aligned} \int_{C_4} \vec{F} \cdot d\vec{r} &= \int_0^1 4t \, dt = \left. 2t^2 \right|_0^1 \\ &= 2 \end{aligned}$$

$$\int_{C_3 \cup C_4} \vec{F} \cdot d\vec{r} = \frac{5}{2} + 2 = \frac{9}{2}$$

Prob 2 : $\delta(x, y, z) = \frac{5}{2} z$

$$\vec{r}(t) = [0, t^2-1, 2t] \quad 0 \leq t \leq 1$$

$$\vec{r}'(t) = [0, 2t, 2] \rightarrow |\vec{r}'(t)| = \sqrt{4t^2 + 4}$$

$$\delta(\vec{r}(t)) = \frac{5}{2}(2t) = 5t$$

$$\text{Mass} = \int_C \delta \cdot ds = \int_0^1 \delta(\vec{r}(t)) \cdot |\vec{r}'(t)| dt$$

$$= \int_0^1 5t \sqrt{4t^2 + 4} dt \quad \begin{array}{l} u = 4t^2 + 4 \\ du = 8t dt \end{array}$$

$$= \frac{5}{8} \int_{t=0}^{t=1} \sqrt{u} du$$

$$= \frac{5}{8} \left(\frac{2}{3} u^{3/2} \right) \Big|_{t=0}^{t=1}$$

$$= \frac{5}{12} (4t^2 + 4)^{3/2} \Big|_0^1$$

$$= \frac{5}{12} [8^{3/2} - 4^{3/2}] = \frac{5}{12} [8 \cdot 2^{3/2} - 8]$$

$$= \frac{10}{3} [2^{3/2} - 1]$$

$$\approx 6.09476$$